Type II conditionals in writings by beginner and advanced English learners: A collostruction-based analysis

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1 Introduction

1.1 Learner corpora in Second langauge acquisition

Collostructional analysis: Stefanowitsch and Gries (2003)

> "For studies that aim to contribute to SLA scholarship by providing insights into the progress of L2 acquisition, it will be important to analyze and compare data from learners at different proficiency levels, ideally following a longitudinal design and using corpora that capture data collected from the same learners at different points in time."

(Römer-Barron, 2026)

conditionals more non-native like (Gries and Wulff, 2021)

1.2 Formulaicity in learner language

beginThese data showed that multimorphemic sequences that go well beyond learners' grammatical competence are very common in early L2 production. Notwithstanding that these sequences contain such forms as finite verbs, wh-questions and clitics, Myles denied this as evidence for functional projections from the start of L2 acquisition because these properties are not present outside chunks initially. Analyses of inflected verb forms suggested that early productions containing them were formulaic chunk (Ellis, 2012)

1.3 Collostructional analysis

The present study is pseudo-longitudinal in nature and intends to observe, through a collostructional lense, aspects of the development of the second conditional in L2 English learners.

RQ1. How are type II conditionals dispersed across learner writing tasks and proficiency level in our analyzed data?

RQ2. Do learners at higher proficiency levels favor different verbs in the protasis of type II conditionals than beginner learners?

RQ3. Does the verbal slot in the protasis of type II conditionals provide evidence for a transition from more formulaic to more productive use of the construction?

2 Methods

2.1 Data: The EFCAMDAT

The EF-Cambridge Open Language Database (EFCAMDAT), first presented by Geertzen et al. (2013), is a large-scale learner corpus containing writings from the online English school *Englishtown*. In their courses, each level corresponds to eight pedagogical units, after each of which students receive a prompt for a writing task that they must complete in order to advance. The corpus comprises responses to these tasks in their original and corrected forms, along with metainformation about the text and the author (learner ID, country of origin, grade given by a teacher, etc.). An advantage of the EFCAMDAT, beyond being generally available for researchers at all stages, is the fact that *Englishtown* levels are aligned with the Common European Framework of Reference (CEFR), making it possible to do a broad range of descriptive analyses of L2 English at globally understood proficency levels.[1]

This study used the version of EFCAMDAT cleaned and post-processed by Öksüz et al. (2025), which contains 620,206 learner writings. However, it was unlikely for second conditionals to be elicited by most prompts; therefore, after qualitatively inspecting the task prompts, I decided on a subset of 20 tasks whose responses were most likely to contain the construction. The selected units and corresponding writing topics are outlined in Table 3 in Appendix A, totaling 33,496 student writings and 4,375,055 words stemming from CEFR levels A2 to C1.

Finally, to ensure sufficiently large samples of both learner texts and type II conditional instances, the writings were grouped into two proficiency bands: learners at CEFR levels A2 and B1 ("lower") and learners at levels B2 and C1 ("upper").

2.2 Extracting type II conditionals

All corrected learner writings in the subcorpus were automatically parsed using Stanza (Qi et al., 2020). Using the corrected texts instead of the original ones is justified by the fact that accuracy does not factor into this study: the goal is not to asses the grammatical proficiency of students, but rather to explore their lexical preferences is a specific slot in different levels. This gives us the ability to use the corrected texts that are present in the corpus, ensuring more robust results of automatic

[1] The availability of large-scale, CEFR-aligned corpora is more limited than one might expect. The most well-known English learner corpus is the Cambridge Learner Corpus (CLC), whose access is highly restricted.

Natural Language Processing (NLP) tools.[2]

The parsed documents were scanned for type II conditionals using both Universal Dependency (UD) relations and part-of-speech (POS) information. Only sentences containing the word "if" as a conditional marker (headed by an `advcl` UD tag, as in Figure 1) and not as an interrogative marker headed by a `ccomp` UD tag, as in Figure 2. Then, the script ensures that the verb in the protasis is in the past tense and the apodosis contains a valid modal verb, thus filtering out other types of conditionals. For this study, I excluded sentences that expressed their conditional value through means other than the grammatical marker "if" (e.g. subject-auxiliary inversion) and sentences containing a modal verb in the protasis (e.g. *If I could go anywhere in the world, I'd go to China.*).

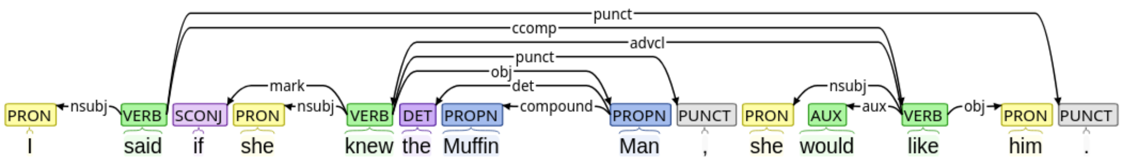

Figure 1: UD parse of a sentence with a conditional *if*

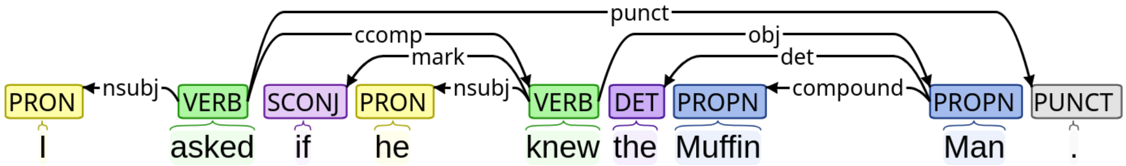

Figure 2: UD parse of a sentence with a non-conditional *if*

Finally, conditionals that fit all requirements were compiled into a CSV file with the full sentence, the lemma filling the verbal slot in the protasis,[3] and relevant metadata about the writing prompt and response, such as the *Englishtown* level the task stemmed from and the topic ID.

2.3 Calculating dispersion across tasks and levels

As a task-based corpus, the texts from EFCAMDAT are subject to task effects (Alexopoulou et al., 2017). That is, the educational environment in which the texts are embedded shapes the grammar and lexis found in the corpus. When drawing conclusions from this kind of corpora, it is possible to run into issues if one assumes all texts to be representative of the general production of learners at their corresponding proficiency, regardless of the task. To observe the effect of task on the frequency of type II conditionals, I applied Gries's (2008) DP (deviation of proportions) measure and its normalized version DP_{norm} for corpus parts of unequal size as presented in the correction by Lijffijt and Gries (2012). The same process was also applied using the binary proficiency levels

[2]However, given previous robustness studies of NLP tools on L2 English (Geertzen et al., 2013), similar results would be expected when running the analysis on the original learner texts.

[3]In the case of constructions with auxiliary verbs, I chose the lemma of the lexical verb as the slot filler in order to avoid artificially inflating the auxiliary's frequency and more accurately reflect lexical variability.

"upper" and "lower" as the corpus parts to ensure that the Distinctive Collexeme Analysis was carried out in comparable parts of the analyzed subcorpus.

2.4 Comparing tendencies across two L2 English proficiency levels

To investigate the differences in verb preferences in the if-clause of second conditionals, I used Distinctive Collexeme Analysis (DCA), an extension collostructional analysis which compares a collexeme's degree of attraction to one construction with its attraction to an alternative, similar construction (Gries and Stefanowitsch, 2004). For the purposes of this study, however, instead of comparing two analogous but distinct constructions, we compare collexemes' association strength to the same constructional slot across two different proficiency levels.

In recent years, Gries has devised a more computationally-efficient way of calculating association strengths in collostructional analysis by replacing contingency table-based computations with the residuals of the χ^2-test, achieving extremely high correlations with the results of the traditional technique (Gries, 2023). Given its efficiency and relative simplicity, the present study adopts this updated approach.

To quantify the lexical diversity of verbs filling the slot under discussion, I opted for an entropy-based approach. Information-theoretic entropy (H) measures the uncertainty in a probability distribution: the more evenly distributed, the higher the entropy. When normalized to a 0-1 scale (H_{rel}), entropy values become comparable across distributions with different numbers of categories (Gries and Ellis, 2015). Entropy-based measures have been used in SLA corpus research to investigate L2 development (for example, see Sun and Wang (2021) for an analysis of entropy and linguistic complexity done on the EFCAMDAT) and was thus deemed an appropriate measure for the purpose of comparing the breadth and distribution of verb lemmas in the relevant position.

3 Results

3.1 RQ1: Dispersion across tasks and proficiency levels

To examine the evenness of occurrence of second conditionals across tasks and proficiency levels, I calculated dispersion indices (DP and DP_{norm}) using different definitions of corpus "parts". First, I partitioned the data into the 20 task topics. Second, the data was divided by proficiency level (lower vs. higher). Finally, I calculated topic dispersion separately within each proficiency group. Table 1 reports DP_{norm}, since raw DP values are nearly identical.

Overall, dispersion across task topics was relatively high, suggesting that different tasks elicited considerably different numbers of type II conditionals in their responses. Nevertheless, when stratifying the tasks by proficiency levels, higher-level tasks showed a more uneven distribution of this

Partitioning / subset condition	$\mathbf{DP_{norm}}$
Topics across all data	0.559
Proficiency level as partitioning variable	0.022
Topic dispersion within proficiency groups	
Low proficiency (topics)	0.182
High proficiency (topics)	0.376

Table 1: Dispersion of type II conditionals. The last block shows topic dispersion calculated separately for low- and high-proficiency subsets.

construction than lower-level ones. On the other hand, dispersion across levels was very low, revealing that both proficiency groups produced a very similar number of conditionals to what would be expected by their corresponding size.

3.2 RQ2: Tendencies in protasis verb choice by proficiency level

Association strengths between the verbal collexemes in the if-clause of second conditionals in writings by higher- and lower-proficiency students were contrasted by means of DCA using the residuals of a χ^2-test. Figure 3 illustrates the results of those verbs with at least five occurrences in the dataset. Positive residuals indicate attraction toward conditionals produced by upper-level learners, whereas negative residuals indicate that the verb was favored by lower-proficiency students.

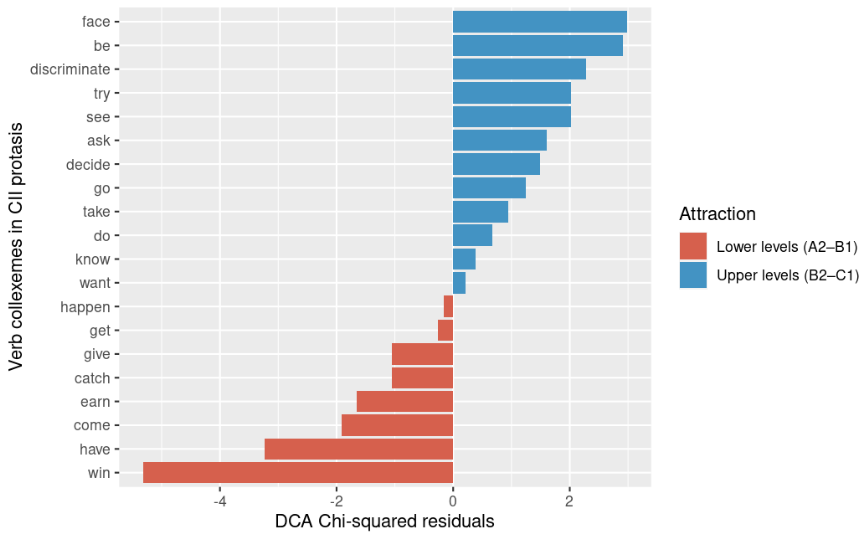

Figure 3: Residuals of the χ^2-test for DCA

The analysis reveals a clear attraction of a few collexemes to one or the other group of students. Advanced students disproportionately favored some higher-level words like "face" or "discriminate", as well as some high-frequency verbs like "be", "try" and "see". On the other hand, the verb "win" was notably overrepresented in texts written by beginners.

3.3 RQ3: Lexical diversity in protasis verbs

Normalized Shannon's Entropy (H_{rel}) was used to measure the diversity of verbs used by beginner and advanced students as a proxy for measuring a potential shift from formulaicity to productivity in type II conditionals. The results are summarized in Table 2 below.

Proficiency level	Entropy (H)	Normalized entropy ($\mathbf{H_{rel}}$)
Lower	2.391	0.628
Higher	2.713	0.639

Table 2: H and H_{rel} values for verb distributions across proficiency groups.

The raw entropy values for the advanced learner texts is slightly higher than for beginner writings. This difference becomes smaller, however, when the entropy values are normalized, suggesting that higher-proficiency students produce overall a larger number of unique verb types, though the evenness of the distribution at both levels is similar.

4 Discussion and conclusion

Observing the high dispersion of type II conditionals across task topics reveals clear task effects: the number of conditionals elicited by each task is indeed not proportional to what would be expected simply from the number of words in their responses. The most likely explanation for this is that some tasks simply require more reasoning about imaginary or unlikely events in the present or future than others. Similar conclusions have been reached regarding the EFCAMDAT not in terms of a specific construction, but with a general focus on different measures of linguistic complexity (Alexopoulou et al., 2017). An alternative reason for an uneven distribution of CII across tasks is the explicit introduction of the construction within specific units, resulting in students being primed to produce a greater number of sentences containing this specific structure. These are not mutually exclusive, but rather, it is to be expected that learning material designers intentionally provide students with opportunities to use newly-introduced grammar means.

On the other hand, perhaps surprisingly, this study reports the opposite pattern regarding student level: second conditionals appear to be produced in similar proportions by students in lower and higher levels of proficiency. It is very likely that this observation is due to the second conditional being acquired at an intermediate level like B1, as suggested by the English Grammar Profile (O'Keeffe and Mark, 2017), and this is where most of the tasks and condtionals from the lower-proficiency level in this study come from. Then, once this construction is available to the learners, its use is maintained in further stages of L2 development, since few alternative structures exist that fulfill the same communicative function.

However, despite the relatively stable overall frequencies observed across proficiency levels, subsequent analyses revealed important differences in the lexical composition of type II conditionals.

The DCA results for higher-proficiency learners included verbs such as *face* (e.g., *face one's fears*) and *discriminate*, which are classified at the B2 and C1 levels, respectively, in the English Vocabulary Profile (EVP) (Capel, 2010, 2012). At the same time, highly frequent and semantically general verbs such as *be*, *try*, and *see* remained among the most strongly associated verbs in the construction.

- Why lower/almost equal entropy in higher levels? -> Maybe specific tasks specially require specific verbs, so that it is more uniform? Could observe dispersion of different words per task and find common words that are unevenly dispersed? There are more words and more conditionals in advanced texts, so it doesnt necessarily mean growth in vocab either. But anyway, would native speakers behave differently? like, obviously they would have a broader vocab, but would it actually have a higher entropy? IDK!!
- Idk if the differences in residuals are significant. For that, should do bootstrapping (Gries, 2023)

Appendices

A Task topics selected for analysis

Table 3 lists all the task topics whose responses were analyzed, along with their corresponding CEFR and *Englishtown* levels.

CEFR level	EF level:Unit	Topic ID	Words	Topic
A2	6:6	46	290,594	Writing an email of advice
B1	8:3	59	474,441	Making a ‘to do’ list of your dreams
B1	9:3	67	354,123	Making a business proposal
B1	9:4	68	284,752	Signing a waiver to go skydiving
B2	10:1	73	797,218	Helping a friend find a job
B2	10:2	74	492,941	Doing a survey about discrimination
B2	10:3	75	438,282	Requesting a bank loan
B2	10:5	77	313,135	Finding a home for a wealthy client
B2	11:2	82	291,016	Helping a coworker deal with a phobia
B2	11:8	88	24,119	Improving your study skills
C1	13:1	97	244,337	Writing a campaign speech
C1	13:4	100	115,523	Giving advice about budgeting
C1	13:8	104	21,223	Reaching your potential
C1	14:2	106	41,642	Choosing a renewable energy source
C1	14:3	107	61,155	Writing a rejection letter
C1	14:4	108	55,637	Attending a seminar on stress reduction
C1	14:5	109	52,968	Talking a friend out of a risky action
C1	14:6	110	39,465	Applying for sponsorship
C1	15:7	119	5,182	Writing about future lifestyles
C1	15:8	120	4,899	Interpreting a prophecy

Table 3: Topics by Level and CEFR

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